Topic: Power series basics

Question: Find the radius of convergence of the power series.

$$\sum_{n=1}^{\infty} \frac{(3x-1)^n}{n^2+n}$$

Answer choices:

A $R = -\frac{1}{3}$

B $R = \frac{1}{3}$

C $R = -\frac{2}{3}$

D $R = \frac{2}{3}$

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Solution: B

First, let's rewrite the series.

$$\sum_{n=1}^{\infty} \frac{(3x-1)^n}{n^2+n}$$

Then applying the ratio test to the power series gives

$$L = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right|$$

$$L = \lim_{n\to\infty} \left| \frac{\frac{(3x-1)^{n+1}}{(n+1)^2+n+1}}{\frac{(3x-1)^n}{n^2+n}} \right|$$

$$L = \lim_{n\to\infty} \left| \frac{(3x-1)^{n+1}}{(n+1)^2+n+1} \left(\frac{n^2+n}{(3x-1)^n} \right) \right|$$

$$L = \lim_{n\to\infty} \left| (3x-1) \left(\frac{n^2+n}{(n+1)^2+n+1} \right) \right|$$

$$L = \lim_{n\to\infty} \left| (3x-1) \left(\frac{n(n+1)}{(n+1)[(n+1)+1]} \right) \right|$$

$$L = \lim_{n\to\infty} \left| (3x-1) \left(\frac{n}{n+2} \right) \right|$$

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$$L = |3x - 1| \lim_{n \to \infty} \left| \frac{1}{1 + \frac{2}{n}} \right|$$

$$L = |3x - 1| \left| \frac{1}{1 + 0} \right|$$

$$L = |3x - 1|$$

To determine where the series converges absolutely, we set this value $L < 1$.

$$|3x - 1| < 1$$

$$3\left|x - \frac{1}{3}\right| < 1$$

$$\left|x - \frac{1}{3}\right| < \frac{1}{3}$$

$$-\frac{1}{3} < x - \frac{1}{3} < \frac{1}{3}$$

$$0 < x < \frac{2}{3}$$

So the interval of convergence is $(0,2/3)$, and the radius of convergence is $R = 1/3$.

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Topic: Power series basics

Question: Find the Maclaurin series representation of the function.

$$f(x) = \frac{3}{1 + 2x^2}$$

Answer choices:

A $\sum_{n=0}^{\infty} (2x^2)^n$

B $\sum_{n=0}^{\infty} 3(2x^2)^n$

C $\sum_{n=0}^{\infty} (-1)^n(2x^2)^n$

D $\sum_{n=0}^{\infty} 3(-1)^n(2x^2)^n$

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Solution: D

Because we already have the Maclaurin series representation of

$$f(x) = \frac{1}{1-x}$$

we can replace x with $-2x^2$.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$\frac{1}{1-(-2x^2)} = \sum_{n=0}^{\infty} (-2x^2)^n$$

$$\frac{1}{1+2x^2} = \sum_{n=0}^{\infty} (-1)^n (2x^2)^n$$

Then multiply the series by 3.

$$\frac{3}{1+2x^2} = \sum_{n=0}^{\infty} 3(-1)^n (2x^2)^n$$

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Topic: Power series basics

Question: Find the first and second derivatives of the power series.

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!}(x-3)^n$$

Answer choices:

A $f'(x) = \sum_{n=1}^{\infty} \frac{1}{(n-1)!}(x-3)^{n-1}$ and $f''(x) = \sum_{n=2}^{\infty} \frac{1}{(n-2)!}(x-3)^{n-2}$

B $f'(x) = \sum_{n=0}^{\infty} \frac{1}{(n-1)!}(x-3)^{n-1}$ and $f''(x) = \sum_{n=0}^{\infty} \frac{1}{(n-2)!}(x-3)^{n-2}$

C $f'(x) = \sum_{n=1}^{\infty} \frac{1}{(n-1)}(x-3)^{n-1}$ and $f''(x) = \sum_{n=2}^{\infty} \frac{1}{(n-2)}(x-3)^{n-2}$

D $f'(x) = \sum_{n=1}^{\infty} \frac{1}{(n+1)!}(x-3)^{n-1}$ and $f''(x) = \sum_{n=2}^{\infty} \frac{1}{(n+2)!}(x-3)^{n-2}$

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Solution: A

To differentiate the power series representation, we can apply power rule for derivatives.

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!}(x-3)^n$$

$$f'(x) = \sum_{n=0}^{\infty} \frac{n}{n!}(x-3)^{n-1}$$

$$f'(x) = \sum_{n=0}^{\infty} \frac{n}{n(n-1)(n-2)(n-3)\ldots}(x-3)^{n-1}$$

$$f'(x) = \sum_{n=0}^{\infty} \frac{1}{(n-1)(n-2)(n-3)\ldots}(x-3)^{n-1}$$

$$f'(x) = \sum_{n=0}^{\infty} \frac{1}{(n-1)!}(x-3)^{n-1}$$

Alternatively, we could have expanded the original power series and differentiated term-by-term,

$$f(x) = 1 + (x-3) + \frac{1}{2!}(x-3)^2 + \frac{1}{3!}(x-3)^3 + \ldots$$

$$f'(x) = 0 + 1 + \frac{1}{2!} \cdot 2(x-3) + \frac{1}{3!} \cdot 3(x-3)^2 + \ldots$$

$$f'(x) = 1 + \frac{1}{1!}(x-3) + \frac{1}{2!}(x-3)^2 + \ldots$$

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and then we can see from the pattern of this expanded series that the power series representation of the derivative is

$$f'(x) = \sum_{n=1}^{\infty} \frac{1}{(n-1)!}(x-3)^{n-1}$$

To find the second derivative, we can apply power rule again.

$$f''(x) = \sum_{n=0}^{\infty} \frac{n-1}{(n-1)!}(x-3)^{n-2}$$

$$f''(x) = \sum_{n=0}^{\infty} \frac{n-1}{(n-1)(n-2)(n-3)(n-4)\dots}(x-3)^{n-2}$$

$$f''(x) = \sum_{n=0}^{\infty} \frac{1}{(n-2)(n-3)(n-4)\dots}(x-3)^{n-2}$$

$$f''(x) = \sum_{n=0}^{\infty} \frac{1}{(n-2)!}(x-3)^{n-2}$$

Alternatively, we could have expanded the power series representation of the first derivative and differentiated term-by-term,

$$f'(x) = 1 + \frac{1}{1!}(x-3) + \frac{1}{2!}(x-3)^2 + \frac{1}{3!}(x-3)^3 + \dots$$

$$f''(x) = 0 + 1 + \frac{1}{2!} \cdot 2(x-3) + \frac{1}{3!} \cdot 3(x-3)^2 + \dots$$

$$f''(x) = 1 + \frac{1}{1!}(x-3) + \frac{1}{2!}(x-3)^2 + \dots$$

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and then we can see from the pattern of this expanded series that the power series representation of the second derivative is

$$f(x) = \sum_{n=2}^{\infty} \frac{1}{(n-2)!}(x-3)^{n-2}$$

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